Dr PUNEET KUMAR MISHRA



Geomorphology | Remote Sensing

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RESEARCH INTERESTS

My research focuses on coastal geomorphology and morphodynamics, emphasising shoreline evolution, coastal landscape change, and geomorphic processes using advanced geospatial techniques. My expertise includes remote sensing and GIS-based shoreline change analysis, embayed beach and estuarine dynamics, DEM/DTM-based coastal terrain analysis, and the assessment of climate change impacts on coastal systems through the integration of satellite and field observations.

EDUCATION

1. **PhD in Science (Earth Science/Marine Science/Remote Sensing)**
   Title: ***Geomorphological studies of the central west coast of India.***
   Academy of Scientific and Innovative Research (AcSIR) | 2021-2026
   Work carried out at CSIR–National Institute of Oceanography, Goa, India
2. **Master of Science in Applied Geology**
   Aligarh Muslim University, India | 2019-2021
3. **Bachelor of Science in Geology, Zoology and Botany**
   Hemvati Nandan Bahuguna Garhwal University, India | 2016-2019

PUBLICATIONS

Peer-Reviewed Journal Articles

1. ***Puneet Kumar Mishra***, Mani Murali R et al.2024, *Investigating the evolution of sand bars and natural channel formation in the Mandovi estuary along the central west coast of India*, Regional Studies in Marine Sciences, Volume 80,2024,103875, ISSN 2352-4855, https://doi.org/10.1016/j.rsma.2024.103875

2. ***Puneet Kumar Mishra***, Mani Murali, R., Deepika Dwivedi et al., 2025.*Impacts of climate change on the morphodynamics of embayed beaches along the Central West Coast of India*. MarGeol 107590. https://doi.org/10.1016/j.margeo.2025.107590.

3. Deepika Dwivedi, Mani Murali R, ***Puneet Kumar Mishra.*** 2026. *Delta under Pressure: A Holistic Assessment of Morphodynamic Change in the Indian Sundarbans from 1972 to 2025*, Marine Pollution Bulletin, Volume 222, Part 1,2026,118653, ISSN 0025326X, https://doi.org/10.1016/j.marpolbul.2025.118653

4. ***Puneet Kumar Mishra***, R Mani Murali, Deepika Dwivedi et al.2026, *Monsoon-Driven Seasonal Variability in Sediment Texture of Embayed Beaches Along the Central West Coast of India*, Regional Studies in Marine Science, 2026, 105155, ISSN 2352-4855, https://doi.org/10.1016/j.rsma.2026.105155.

5. Dinesh Kumar Sahu, ***Puneet Kumar Mishra*** et al.2026**,** *Impact of Seasonal Wave Hydrodynamics on Beach Morphology and Sediment Grain Size along the Central West Coast of India.* Regional Studies in Marine Science, Volume 101,2026,105324, ISSN 2352-4855, https://doi.org/10.1016/j.rsma.2026.105324.

6. Ashutosh Gairola, Poonam Srivastava, Dipti Joshi, Manoj Joshi, Jitesh Singh Bisht, ***Puneet Kumar Mishra***. Spatial Diversity and Abundance of *Bactrocera* spp. Fruit Flies in the Northwestern Himalaya of Uttarakhand Using Methyl Eugenol Traps. *Applied Fruit Science* **68**, 117 (2026). https://doi.org/10.1007/s10341-026-01832-1

Manuscripts Under Review

1. ***Puneet Kumar Mishra***, Mani Murali, R., Dwivedi, D., "*Millennial-Scale Evolution of India's Central West Coast from Satellite-Derived Morphological Indicators".*

2. Deepika Dwivedi, Mani Murali, R., ***Puneet Kumar Mishra***, Shincy Francis, Santhosh Kumar S, "*Satellite-Based Assessment of Suspended Sediment Dynamics in the Indian Sundarban Delta (1988–2024) and Its Geomorphic Evolution*".

3. SK Ariful Hossain, Mani Murali, R., Shincy Francis, ***Puneet Kumar Mishra*** et al., *Seasonal and annual morphological changes of Rameswaram Island, Gulf of Mannar, Southeast coast of India*.

4. Poonam Srivastava, Sachin Chavan, P.V. Rami Reddy, Sandeep Singh, A. Y. Munj, Vaishali Zote, A. Nithish, Suganya Kanna, ***Puneet Kumar Mishra*** et al., *Decadal epidemiological surveillance and spatial risk assessment of Citripestis eutraphera (Meyrick) across Indian mango ecosystems.*

CONFERENCE / SYMPOSIUM PRESENTATIONS

1. ***Puneet Kumar Mishra***, R. Mani Murali, "*Coastal processes and dynamics of sandbars at Mandovi estuary in the central west coast of India*", Indian Geophysical Union,60th Annual convention on Advances in geosciences with special reference to coastal hazards, 22-24 November 2023.

2. ***Puneet Kumar Mishra***, R. Mani Murali, Deepika Dwivedi, "*Morphological dynamics and climatic influences on embayed beaches along the central west coast of India*", OSICON-2025 9th National Conference of Ocean Society of India, 5-7 February 2025.

3. Deepika Dwivedi, R. Mani Murali, ***Puneet Kumar Mishra***, "*Shoreline Change Analysis Using Remote Sensing: Insights From The Sundarban Delta, India*" OSICON-2025 9th National Conference of Ocean Society of India, 5-7 February 2025.

4. ***Puneet Kumar Mishra***, Mani Murali R, Deepika Dwivedi, "*Shifting Shores: Integrating Remote Sensing and In-situ Observations to Understand Embayed Beach Dynamics under Climate Change along Central West Coast of India",* American Geophysical Union-2025, New Orleans, LA, 15-19 December 2025

5. Deepika Dwivedi, Mani Murali R, ***Puneet Kumar Mishra*** , "*Multi-Decadal Shoreline Dynamics and Morphological Transformation in the Indian Sundarban Delta: A Remote Sensing-Based Assessment (1972–2025)"*, American Geophysical Union-2025, New Orleans, LA, 15-19 December 2025

6. ***Puneet Kumar Mishra***, Mani Murali, R. Dwivedi, D. et al., "*Assessing The Morphodynamic Response Of Embayed Beaches To Climate Change Along The Central West Coast Of India",* World Ocean Science Congress-2026, CSIR-NIO, Goa, India, 23-26 February 2026.

7. Dwivedi, D., Mani Murali, R., ***Puneet Kumar Mishra, "*** *Morphodynamic Shifts in the Indian Sundarban Delta: Tracing the Impact of Natural Processes and Human Interventions (1972–2025)"*, World Ocean Science Congress-2026, CSIR-NIO, Goa, India, 23-26 February 2026.

8. ***Puneet Kumar Mishra,*** Mani Murali, R. Dwivedi, D. " *Decadal to Millennial Evolution of coastline along the Central West Coast of India: Integrating Field Observations, Remote Sensing, and Paleo shoreline Proxies*" EGU 2026, Vienna, Austria, 03-08 May 2026.

9. Dwivedi, D., Mani Murali, R., ***Puneet Kumar Mishra***, "*Exploring Sediment–Morphodynamic Coupling in the Evolving Indian Sundarban Delta"* EGU 2026, Vienna, Austria, 03-08 May 2026.

10. ***Puneet Kumar Mishra,*** Mani Murali, R., Dwivedi, D., "*Morphodynamic Evolution of Embayed Beaches Along the Central West Coast of India Under Changing Climate",* OASIS-2026, CSIR-NIO, Goa, India, 18-19 August 2026.

11. ***Puneet Kumar Mishra,*** Mani Murali, R., Dwivedi, D. "*Dynamic Estuarine Sandbars As Natural Coastal Infrastructure: Implications For Navigation, Coastal Resilience, And The Blue Economy*" American Geophysical Union-2026, San Francisco, CA, 07-11 December 2026 *

12. Dwivedi, D., Mani Murali, R., ***Puneet Kumar Mishra et al.*** *"Suspended Sediment Dynamics and Shoreline Evolution in the Indian Sundarban Delta: Insights from Satellite Observations"* American Geophysical Union-2026, San Francisco, CA, 07-11 December 2026*

**Upcoming*

TECHNICAL EXPERTISE

1. **Remote Sensing:** Image classification, change detection, band ratioing, pan-sharpening, time series analysis, Mineral exploration, coastal zone monitoring, environmental impact assessment (EIA)
2. **GIS analysis:** Change analysis, terrain and geomorphological mapping, vulnerability assessment, LULC mapping, morphometric analysis, GIS analysis, JupyterNotebook, Python, spatial database, geospatial modelling, geoprocessing, geostatistics, raster analysis, vector analysis, geospatial automation
3. **Tools:** ArcGIS Desktop, ArcGIS Pro, CoastSat, ERDAS Imagine, QGIS, ENVI, SNAP, Global Mapper, Google Earth Pro, Google Earth Engine (GEE), MS Office
4. **Field Instruments:** GPS, Optical Level, LISST 200x submersible particle size analyser, CTD, River ADCP, GNSS-RTK (Real-Time Kinematic), OBS 5+ (Optical Backscatter Sensor)

FELLOWSHIPS, AWARDS & HONORS

1. EGU Early Career Scientist Travel Support 2026
2. AGU Student Travel Grant 2025
3. GATE 2025 | (Geology/Geophysics)
4. CSIR-UGC NET JRF June 2020 | (All India Rank-23, Earth, Atmospheric, Ocean and Planetary Sciences)
5. UGC-NET Assistant Professorship, June 2020 | (Geography)

RESEARCH EXPERIENCE

1. Senior Project Associate

01st September 2026 –Present

📍 **Ocean Engineering Division, CSIR–National Institute of Oceanography, Goa, India**

1. **Coastal Geomorphology & Process Analysis:** Conducting advanced research on coastal geomorphological evolution, shoreline dynamics, sediment transport, and coastal hazards by integrating remote sensing, GIS, field observations, and oceanographic datasets.
2. **Geospatial Analysis & Coastal Monitoring:** Designing and implementing multi-temporal geospatial analyses using satellite imagery, DEMs, UAV data (where applicable), and GIS techniques for monitoring shoreline change, coastal landforms, and morphodynamic processes.

3. **Spatial Data Management:** Developing and managing geospatial databases, performing quality assurance, spatial analyses, and process-based interpretation to support coastal engineering, environmental management, and scientific investigations.
4. **Research Project Execution & Technical Coordination:** Supporting the planning, execution, and reporting of multidisciplinary research projects through technical coordination, data integration, progress monitoring, and preparation of technical deliverables.
5. **Scientific Communication & Collaboration:** Preparing high-quality technical reports, peer-reviewed publications, project documentation, and conference presentations while collaborating with interdisciplinary research teams and external stakeholders.

2. Senior Research Fellow *01st September 2023 –31st August 2026*

📍 **Remote Sensing and GIS lab, Physical Oceanography Division, CSIR–National Institute of Oceanography, Goa, India**

1. **Morphodynamic Analysis:** Integrated remote sensing, GIS, and oceanographic datasets to investigate the influence of waves, tides, and sea-level rise on coastal morphology along the central west coast of India.
2. **Geospatial Mapping & Monitoring:** Executed multi-temporal satellite data analyses for mapping and monitoring coastal geomorphology, shoreline dynamics, and sedimentary landforms using advanced GIS and remote sensing techniques.
3. **Data Management & Visualisation:** Developed and maintained spatial databases, performed data quality control, and generated thematic maps and graphical outputs for technical reports and research dissemination.
4. **Scientific Writing & Publication:** Contributed to manuscript preparation, publication in peer-reviewed journals, and presentation of research outcomes at scientific conferences and workshops.
5. **Project Execution & Collaboration:** Supported the execution of research projects through effective coordination, data compilation, and preparation of technical documentation in collaboration with interdisciplinary teams.

3. Junior Research Fellow *24th August 2021 – 31st August 2023*

📍 **Remote Sensing and GIS lab, Physical Oceanography Division, CSIR–National Institute of Oceanography, Goa, India**

1. **Geospatial Data Processing:** Acquired and pre-processed multi-source satellite datasets (Landsat, Sentinel, Resourcesat) for baseline mapping and geomorphic classification of coastal environments.

2. **Topographic & Bathymetric Analysis: Utilised DEMs, tidal data, and coastal profiles to evaluate e**levation dynamics, sediment transport pathways, and nearshore morphological adjustments.
3. **Field Data Acquisition & Integration:** Assisted in ground-truthing, beach profiling, and in-situ data collection, integrating field observations with remotely sensed data for validation and accuracy assessment.
4. **Spatial Statistics & Temporal Analysis:** Implemented geospatial and statistical tools to analyse coastal change drivers, identify geomorphic hotspots, and model coastal response to natural and anthropogenic influences.

RESEARCH PROJECTS

1. Study on shoreline changes off Sikka during and after construction of Berth A2, Gujarat, India
2. Scientific studies to identify the reasons for the sea erosion and suggest suitable measures to regain the beach within the mining lease areas of IREL, OSCOM, Indian Rare Earths Limited, Orissa, India
3. Implementation of Coastal Management Information System (CMIS) in the state of Goa and the region of South Maharashtra
4. A survey in property bearing Sy. 111/1 of Arossim, Goa, India
5. A survey in property bearing 134/3 of Tambori Ward, Cavelossim, Salcette, Goa, India
6. Baseline study for assessing the biodiversity and biophysical status of coral reefs along the Maharashtra coastline: prospects
7. Beach carrying capacity study for the coastal beaches of Goa state, India
8. Shoreline Morpho-dynamics and Coupled Hydro-Morphological Simulation using CWC's CMIS and multi-source observational data for evolution of site-specific Climate-Resilient Protection Plans.
9. Preparation of Management Plans for the Ecologically Sensitive Areas (ESAs) and Islands of the state of Goa, India.

PROFESSIONAL TRAINING & CERTIFICATIONS

1. An Elite certificate in NPTEL online certification 12-week course on **"Remote sensing essentials"** (January -April,2025)

2. Elite+Gold certificate in NPTEL online certification 12-week course on **"Geographic Information System"** (January -April,2024).

3. SWAYAM online course certification 15-week course on **"Basics of Remote sensing GIS & GNSS and its application"** (January-April,2024)

4. Short course on **"Advanced SAR data processing for ground surface movement",** Oct 14-25,2024 at IIRS, Dehradun.

5. 22nd course on **"Application of Remote Sensing and GIS in Mineral Exploration"** under the National Natural Resource Management System (NNRMS) program of ISRO conducted by PGRS Division, Geological Survey of India Training Institute, Hyderabad from 01.05.2024 to 21.05.2024.

REFERENCES

1. **Prof. Jaya Kumar Seelam**
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2. **Mr Ilangovan D**
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